

Dr eman gomaa

Consultant dermatology

Urticaria

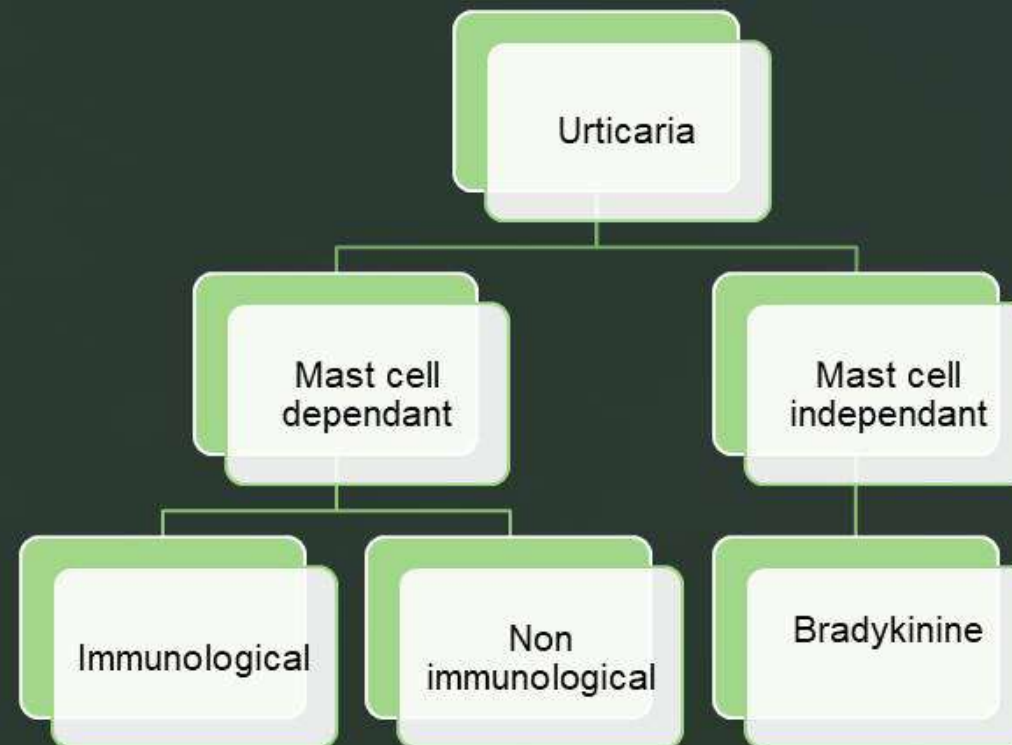
Urticaria



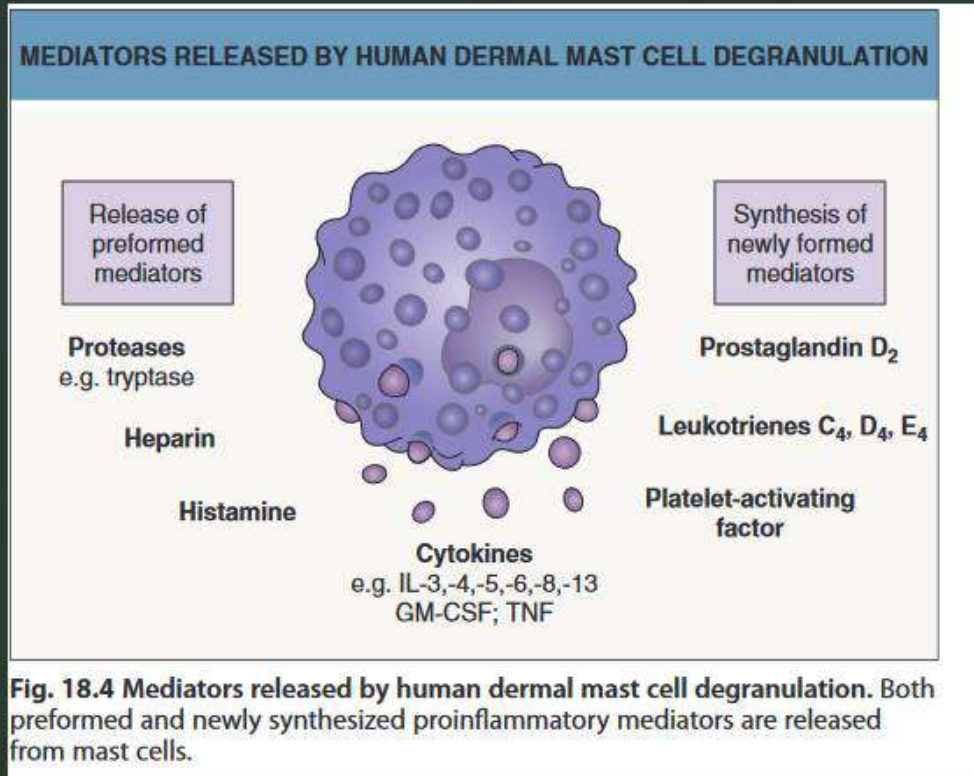
Urticaria

- Disease spectrum of wheal or angioedema or both
- According to depth of swelling
 - Wheal :pink swelling in superficial dermis
 - Angioedema : swelling at deep dermis ,sc

aetiopathogenesis



Mast cell



- Is the main effector of urticaria
- Distrebuted all over body
 - 2 types :
 - Skin , intestinal mucosa >>>> tryptase , cymas
 - Bowel mucosa , alveolar , nasal >>>> tryptase only
- Expree high affinity to igE receptor

MAST CELL DEGRANULATING STIMULI

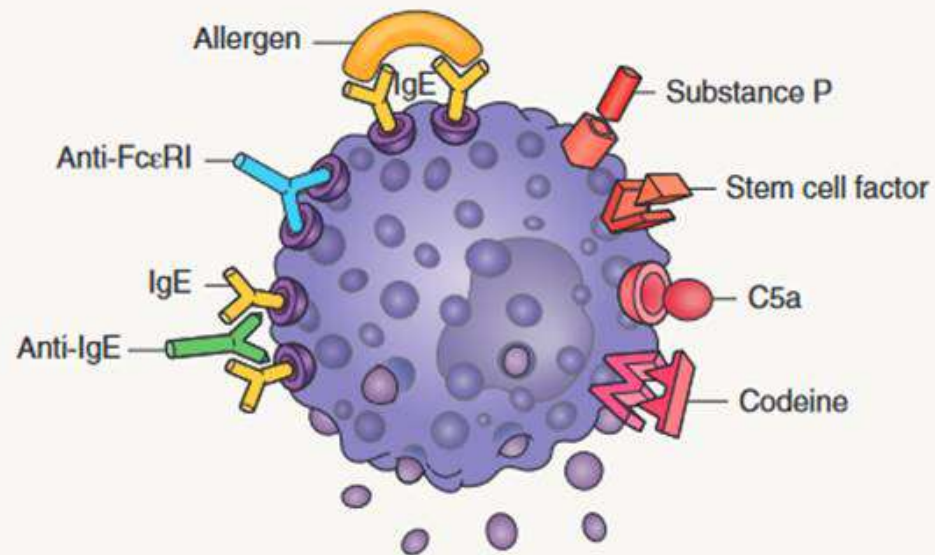
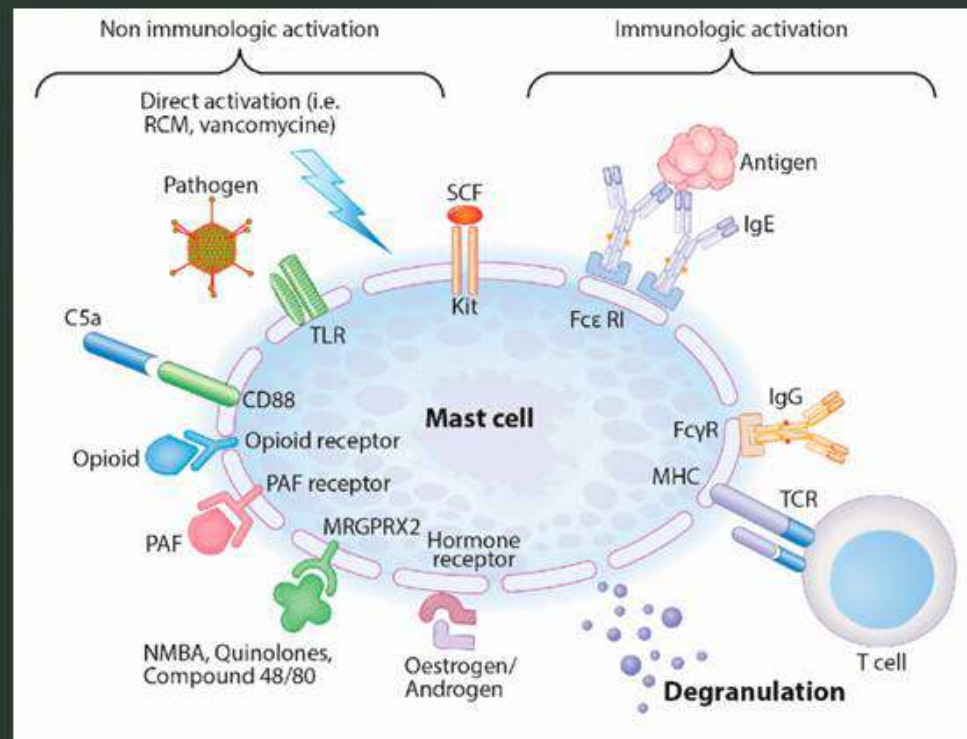


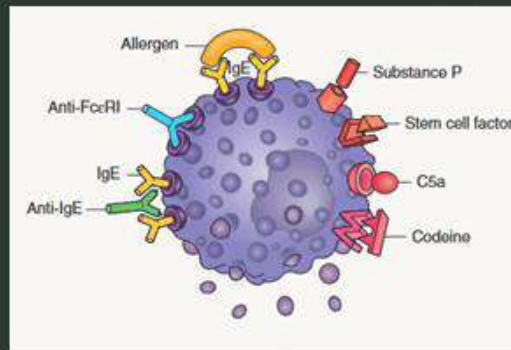
Fig. 18.3 Mast cell degranulating stimuli. Both immunologic and non-immunologic stimuli can lead to release of mediators. Stem cell factor is also known as KIT ligand.



Mast cell dependant

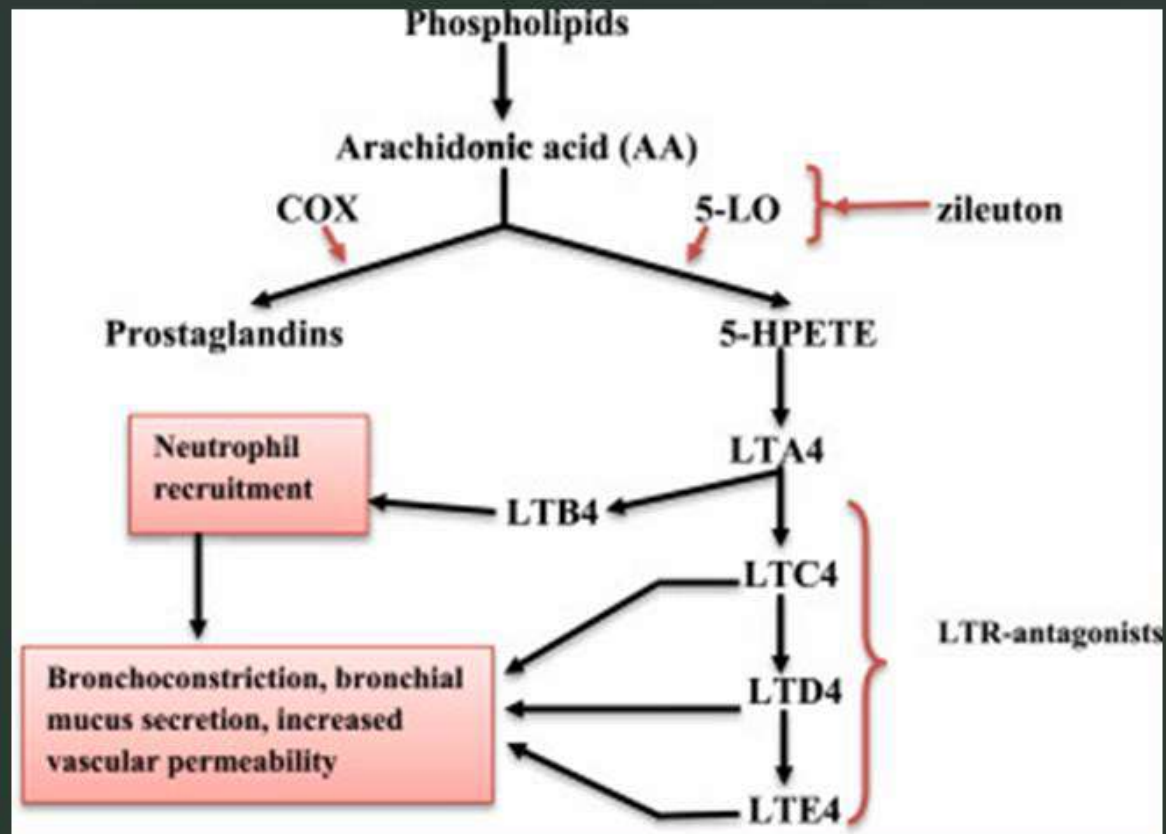
Immunological

1. Cross linking of IgE receptor to allergen
2. IgG autoantibody against IgE
3. Anti Fc receptor



Non immunological

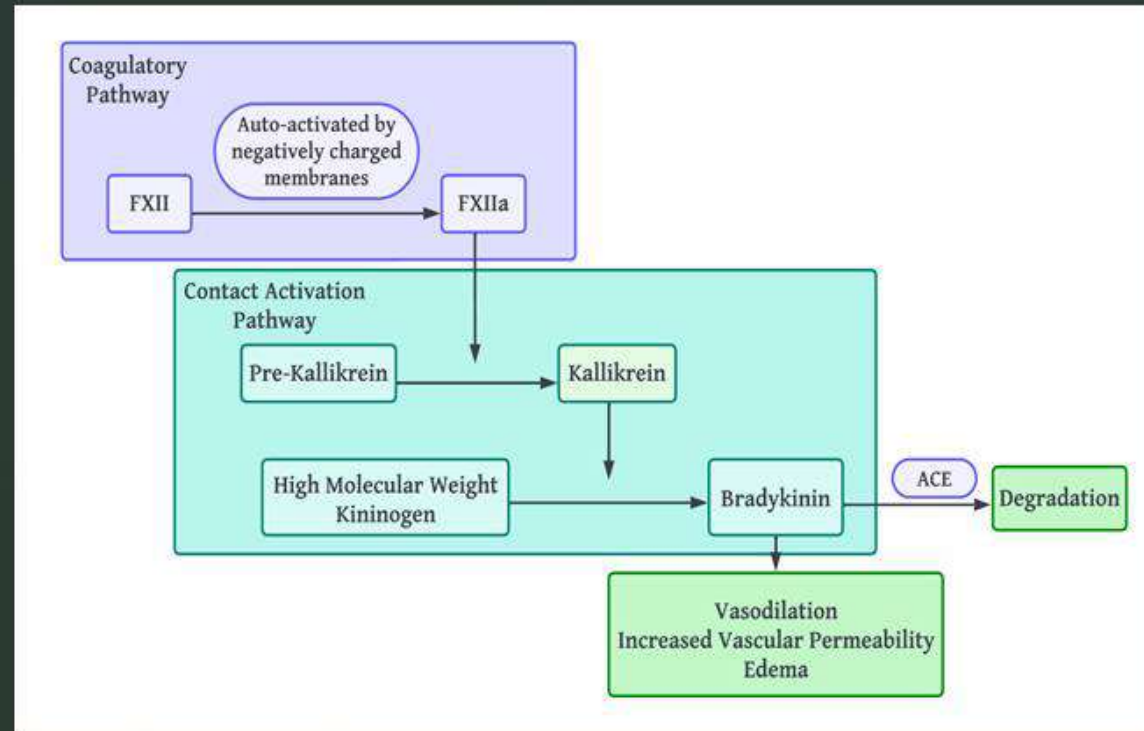
1. Direct mast cell releasing agent (C5a, codeine, opiates, substance P, stem cell factor)
2. Diversion of PG >>> leukotriene
NSAID, food additive

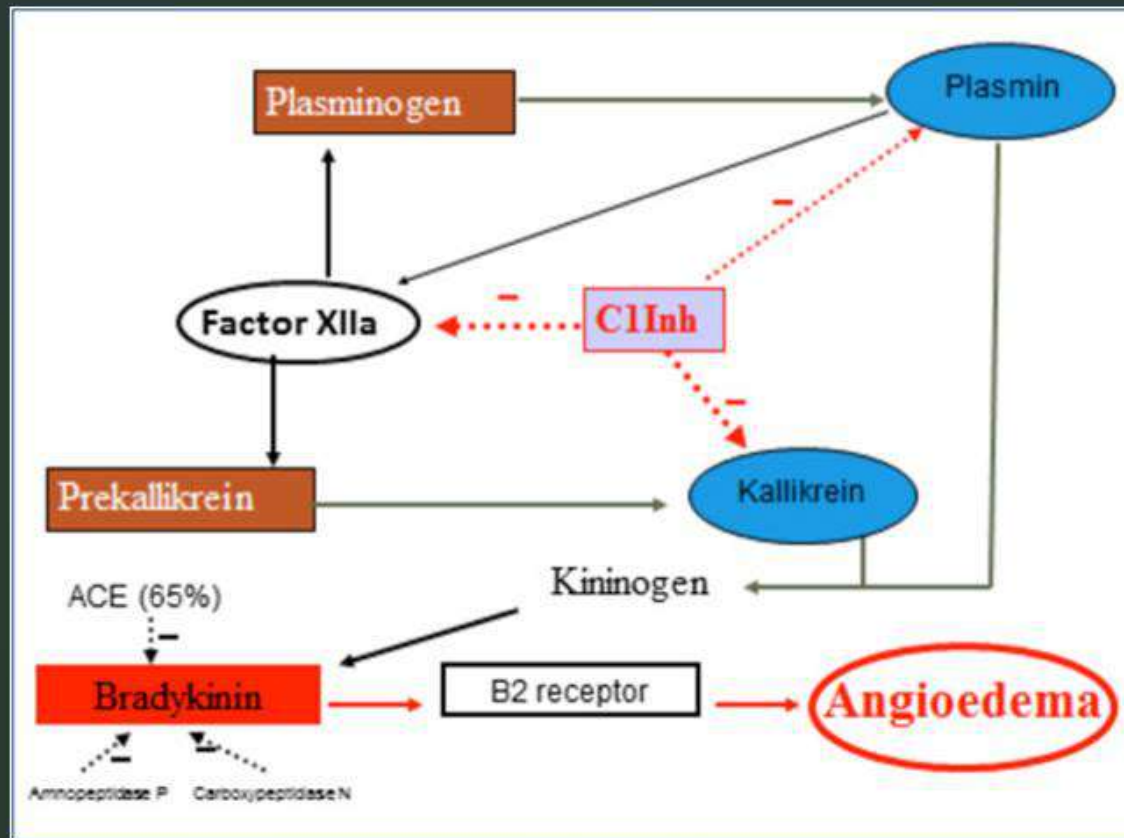


Mast cell independent



1.  C1 esterase inhibitor
2. Activating mutation in hagemen factor
3. Ace inhibitor





Classification of urticaria

angioedema

Urticarial
vasculitis

syndroms

Physical

contact

Spontaneous urticaria

URTICARIA – CLINICAL CLASSIFICATION AND DIFFERENTIAL DIAGNOSIS

Clinical classification

- Spontaneous urticaria (all urticarias not classified below)
- Inducible urticarias (see [Table 18.3](#))
 - Physical and cholinergic urticarias
 - Contact urticaria (induced by percutaneous or mucosal penetration)*

Differential diagnosis

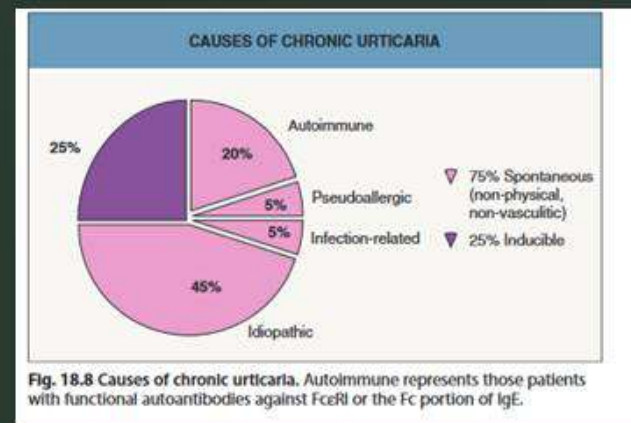
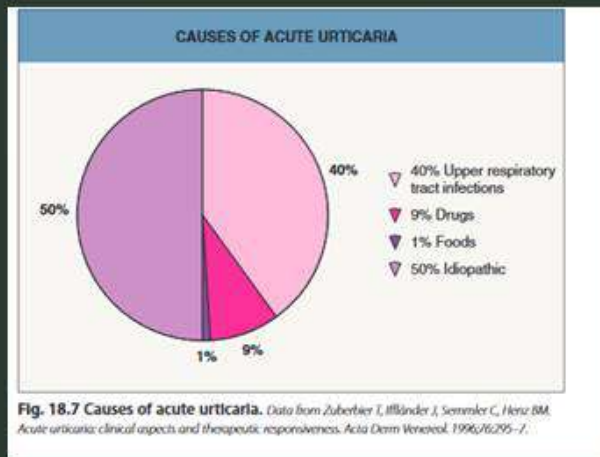
- Urticarial vasculitis (defined histologically by leukocytoclastic vasculitis)
- Angioedema without wheals (e.g. hereditary angioedema, ACEI-induced angioedema)
- Distinctive urticarial syndromes (e.g. autoinflammatory syndromes)

*Included within the inducible urticarias in the EAACI/GA²LEN/EDF/WAO Guideline⁵⁵.

Spontaneous urticaria

acute

chronic



Acute spontaneous urticaria

- Common in young children with atopic dermatitis
- Multiple pruritic wheals of different sizes erupt anywhere on the body and then fade within 2–24 hours without bruising
- angioedema may last up to 72 hours when severe
- Systemic symptoms of fatigue, sweats and chills, indigestion, myalgia or arthralgia may occur with severe attacks
- pyrexia or arthritis should alert the clinician to another explanation, such as urticarial vasculitis, Schnitzler syndrome, or cryopyrin-associated periodic syndrome (e.g. Muckle–Wells syndrome).

Chronic spontaneous urticaria

- At least twice a week for more than 6 weeks
- 4th decade
- associated with autoimmune thyroid, H. pylori
- Parasitic infection intestinal strongyloidiasis

Physical (inducible) urticarias

CLASSIFICATION OF INDUCIBLE URTICARIAS

Urticaria due to mechanical stimuli

Dermographism

- Immediate
 - Simple
 - Symptomatic
- Delayed

Delayed pressure urticaria

Vibratory angioedema

- Inherited
- Acquired

Urticaria due to temperature changes

Heat

- Heat contact urticaria

Cold

- Cold contact urticaria
 - Primary
 - Secondary (cryoproteins)

Urticaria due to sweating or stress

Cholinergic urticaria

Adrenergic urticaria

Exercise-induced urticaria

- Heat contact urticaria
- Cold contact urticaria
- Exercise-induced anaphylaxis
- Food- and exercise-induced anaphylaxis

Contact urticaria

Solar urticaria

Aquagenic urticaria

Rare variants are not shown here for simplicity (see text).

mechanical



Fig. 18.9 Symptomatic immediate dermographism. Itchy wheals appear very soon after scratching. *Courtesy, Jean I. Bolognia, MD.*



Fig. 18.10 Delayed pressure urticaria.

- Sharply localized wheal within second after skin strocking



Fig. 18.11 Cold urticaria. Wheals developed on the forearm after placement of an ice cube for 10 minutes, followed by rewarming. *Courtesy, Thomas Schwartz, MD.*



Fig. 18.12 Cholinergic urticaria on the trunk after hot bath provocation.



Figure 38-6 Lesions of cholinergic urticaria observed in a patient after 15 minutes of exercise in a warm room.



Fig. 22.9 Aquagenic urticaria on the back after swimming, showing a few small papular weals surrounded by wide flares. (Courtesy of Norfolk and Norwich University Hospital, UK.)

◇ May be **due to water soluble antigen in the horny layer** which diffuses into the dermis and activate mast cell

Contact urticaria

- Occur when substance is applied to intact skin

Allergic (immunological)

less common
Type 1 hypersensitivity
latex

Non allergic
(Non immunological)
No prior sensitization
Due to PG2
Not histamin
Cosmetic , food additive

Urticarial vasculitis

- Type of csv (leukocytoclastic vasculitis)
- Wheal :
 - Persist more than 24 h
 - Purpra
 - Painfull
 - Hyperpigmentation
- Type 3 hypersensitivity



Fig. 18.13 Urticarial vasculitis. Lesions look like spontaneous urticaria but last longer and may bruise. An incidental finding is a skin graft donor site.

Pathology: Leucocytoclastic vasculitis

1. Endothelial cell swelling
2. Fibrinoid necrosis of blood vessel wall
3. Inflammatory infiltrate within and around dermal blood vessels
4. Neutrophils with fragmented nuclei (Karyorrhexis)
5. Extravasation of RBCs
6. In late cases: Thrombosis of blood vessel

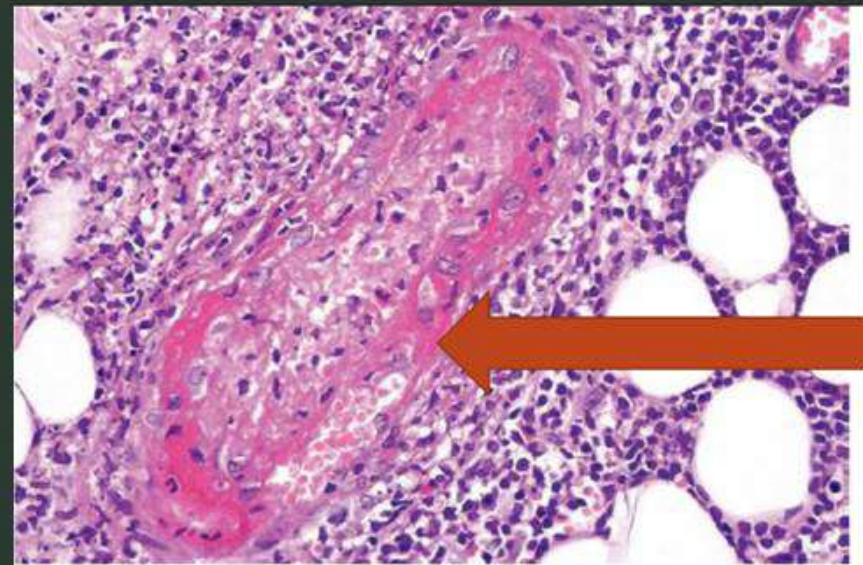


Fig. 16.16

Leukocytoclastic vasculitis: high-power view showing fibrinoid necrosis and a mixed inflammatory cell infiltrate composed of neutrophils, eosinophils, and lymphocytes. There is marked leukocytoclasia (karyorrhexis, nuclear dust).

URTICARIAL VASCULITIS	
Epidemiology	Favors middle-aged women
Characteristics of urticarial lesions	>24 hours in duration; painful & burning sensation as well as pruritus; residual purpura as they resolve
Sites of involvement	Often occur at pressure points
Concurrent angioedema	Up to 40% of pts
Extracutaneous manifestations (% of pts affected)*	Arthralgias (50%) – transient, migratory GI involvement (20%) – abdominal pain, nausea, vomiting, diarrhea Pulmonary obstructive disease (20%) Renal (5–10%) – proteinuria, hematuria Ocular (unusual) – conjunctivitis, episcleritis, uveitis Other (unusual) – Raynaud phenomenon, livedo reticularis, splenomegaly, lymphadenopathy, idiopathic intracranial HTN, pericardial or muscle involvement
Associated disorders	Systemic lupus erythematosus Sjögren syndrome Serum sickness Cryoglobulinemia Infections – hepatitis B or C virus, Epstein-Barr virus Rarely, solar or cold urticaria, drugs, hypergammaglobulinemia
Laboratory evaluation	CBC with differential, platelet count; ESR, CRP; LFTs, BUN, creatinine Complement assays – C3, C4, CH50; if C4 low, C1q Urinalysis If indicated, pulmonary function tests, CXR Tests for associated disorders, e.g. ANA, anti-Ro Ab
Disease course	Average of 3 years
Treatment (primarily based upon small case series and case reports; see Table 24.10)	Antihistamines for associated pruritus NSAIDs – improvement in 50% of patients Colchicine, dapsone, hydroxychloroquine, methotrexate, plasmapheresis Corticosteroids ± immunosuppressants†
*Higher incidence and greater severity in patients with hypocomplementemia. †Especially for hypocomplementemic urticarial vasculitis.	

Angioedema



angioedema

1- hereditary

Type 1
↓
Level of C1 inhibitor

Type 2
↓
Function of C1 inhibitor

Type 3
Activating mutation of f12
hageman

↓
Acquired
C1inh

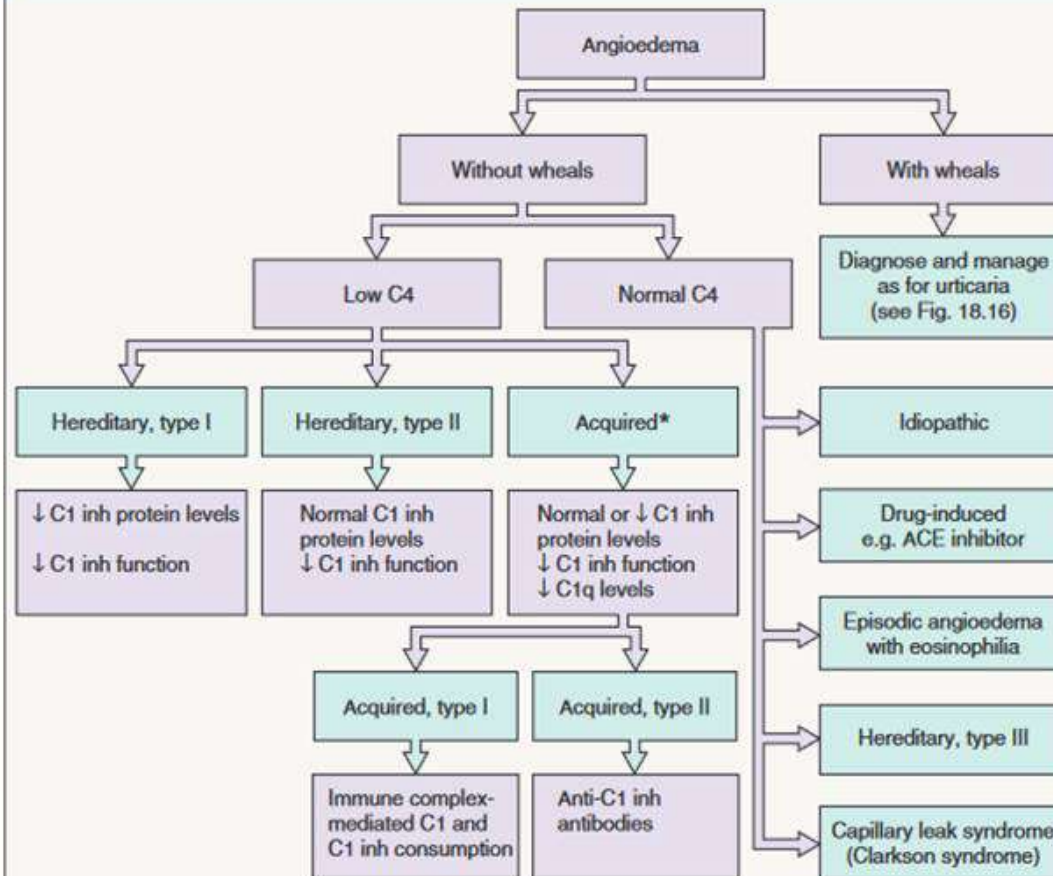
B cell lymphoma
Plasma cell dyscrasis
ACTD >>> SLE
C1 inh Ab

Drug

ACEI
NSAID
Estrogen

idiopathic

ALGORITHM FOR THE DIAGNOSIS OF ANGIOEDEMA



*ACE = Angiotensin-converting enzyme inhibitor therapy

Hereditary Angioedema

- ◆ Type 1: mutation resulting in reduced **levels** of C1 esterase inhibitor
- ◆ Type 2: mutation resulting in reduced **function** of C1 esterase inhibitor
- ◆ Type 3: **activating** mutation in gene that encodes FXII (Hageman factor)

Increased Bradykinin

❖ Suspect type 1 and 2:

1. If there is a family history
2. Previous laryngeal edema or colicky abdominal pain
3. The usual duration is 48–72 hours, and episodes are often followed by a refractory period

- ❖ Type 3: the age of onset is later usually in the second decade and there is a higher frequency of facial angioedema

NB:

- ◊ In type 1 and 2, there is deficiency of C1 inhibitor leading to activation of C1 and activation of C4 (C4 level in serum is decreased) (Normal in type 3)
- ◊ ACEI inhibit ACE (bradykininase) leading to increased bradykinins and angioedema (most cases develop within 3 weeks of treatment but may be seen after one year)
- ◊ Acquired C1 inhibitor deficiency occurs in patients with B cell lymphoproliferative disorders, plasma cell dyscrasia and autoimmune CT diseases (C4 level in serum is decreased)

Investigation for angiedema

1. C4 level >>>
 1. Low in type 1 – 2 HAE , acquired angioedaema
 2. Serum C1q : dcrease in acquired ae

■ treatment of angioedema

C1 Esterase Inhibitor (C1 inh) Deficiency

Emergency:

- Intravenous C1 inh concentrate
- Or **Fresh frozen plasma** may be given as an alternative to C1 inh concentrate
- Or **subcutaneous Icatibant**
(a specific bradykinin B2 receptor antagonist)
- Or subcutaneous Ecallantide (a kallikrein inhibitor)

For minor surgical procedures:

- **Oral tranexamic acid** for 48 hours before and after the procedure

For Maintenance:

- Only necessary for patients with symptomatic recurrent angioedema or related abdominal pain
- **Danazol** (needs to monitor virilizing side effects, hepatic, adenomas)

Urticarial syndrome

1. Cryopyrine associated periodic syndrome (CAPS)



Familial cold auto-inflammatory syndrome (FCAS)
Milder form of CAPS
Causes skin rash, mild fever and arthralgia within a few hours of cold exposure



Muckle-Wells syndrome (MWS)
Most common form of CAPS with episodes of rash, fever, red eyes, arthralgia and fatigue
Progressive sensorineural hearing loss is common



Chronic infantile neurologic, cutaneous and articular (CINCA)
Most severe form of CAPS
Severe neurological manifestations, arthropathy, fever and rash



Schnitzler syndrome

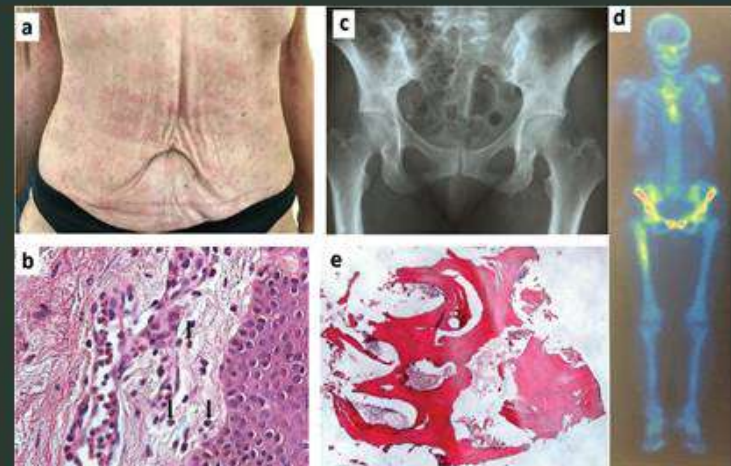
Major criteria

- Urticarial skin rash
- Monoclonal IgM component or IgG by variant type

Minor criteria

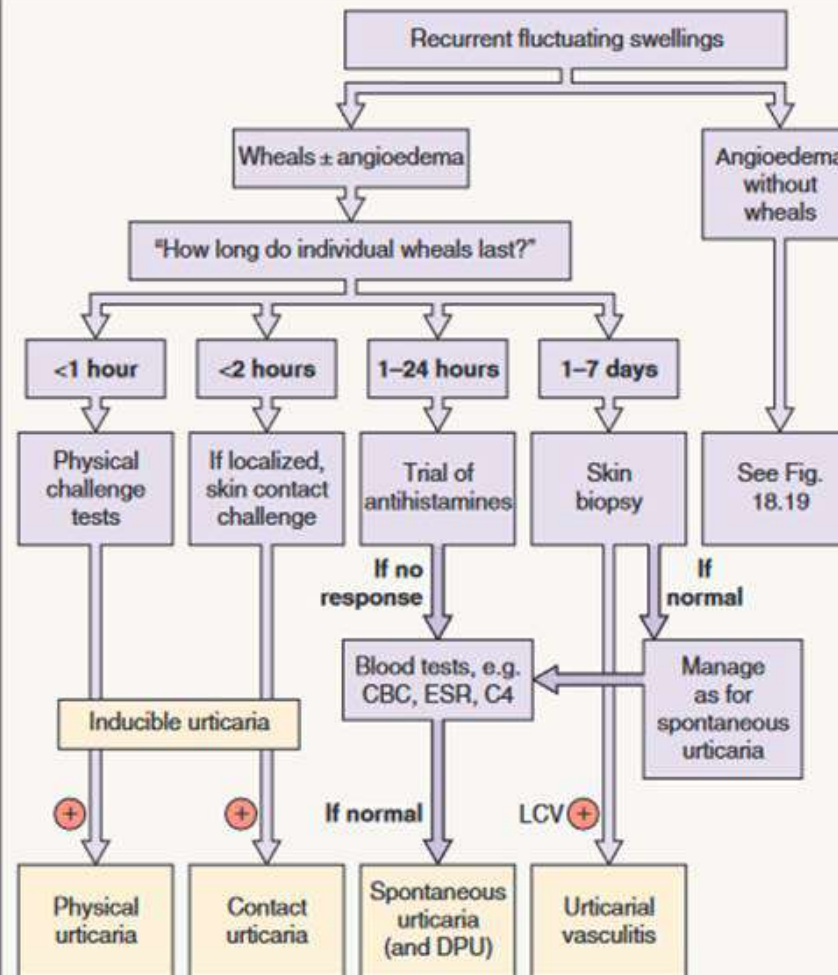
- Intermittent fever or subfebrile fever episodes
- Arthralgia or arthritis or bone pain
- Palpable lymph nodes
- Liver or spleen enlargement
- Elevated erythrocyte sedimentation rate
- Leukocytosis
- Abnormal finding on bone morphologic investigation

Diagnosis of schnitzler's syndrome requires the presence of two major criteria and at least two minor criteria



- 
- 3- episodic angioedema with eosinophoilis
 - 4- systemic capillary leak syndrome

APPROACH TO THE DIAGNOSIS OF CHRONIC URTICARIA



CBC = Complete blood count
LCV = Leukocytoclastic vasculitis

DPU = Delayed pressure urticaria

Evaluation

History

Duration
Frequency
Duration of individual lesion
Associated illness
Family history
Occupation
Drug

Clinical examination

Morphology of lesion
Duration of lesion
Sign of systemic

Investigation

Acute spontaneous urticaria
IGE against the allergen

Skin prick test
Serum radioallergen sorbent test (RAST)

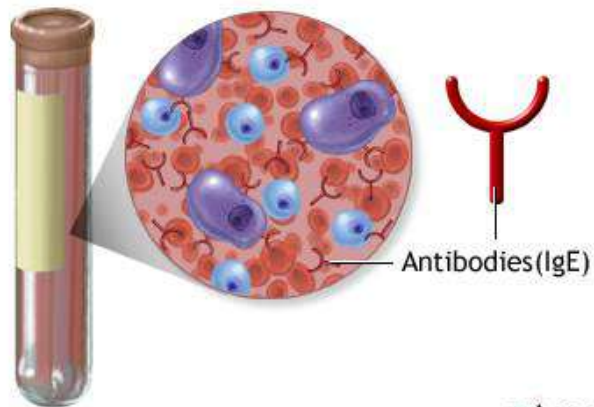
Chronic spontaneous urticaria
Cbc
Urin analysis
Stool analysis
Thyroid function
Immunoglobulin

Physical urticaria
Provocative skin test

Cold
Dermatographism
Aquagenic
Cholinergic
Delayed pressure
Solar
Localized heat

Autoimmune urticaria
ASST
Detect circulating Ab to IgE receptor

The blood test measures the levels of allergy antibody, or IgE, produced when your blood is mixed with a series of allergens in a laboratory



ADAM.

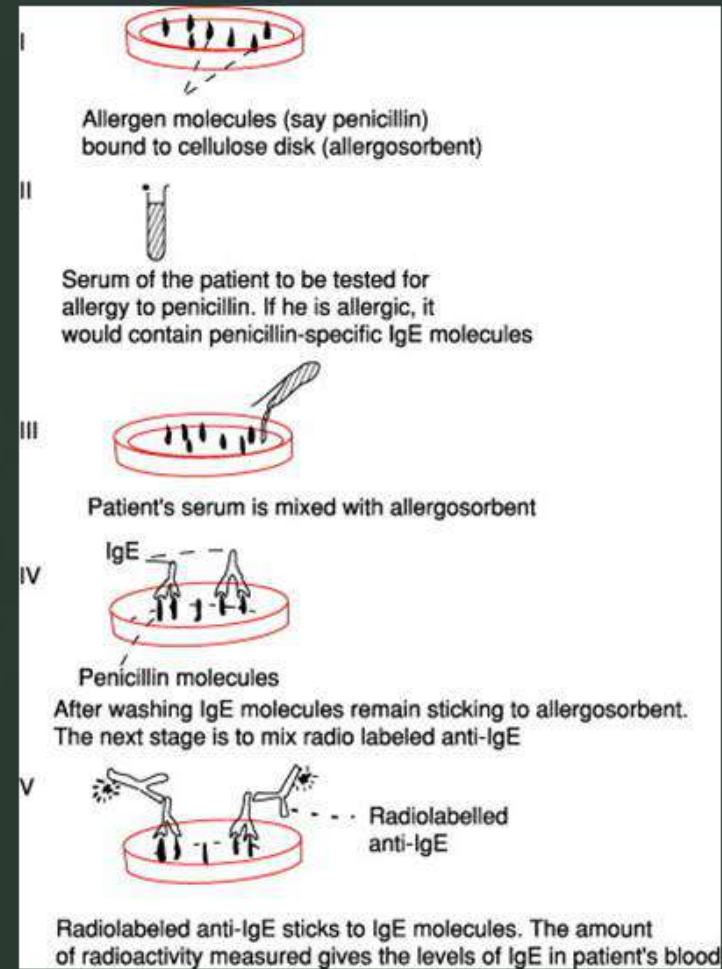
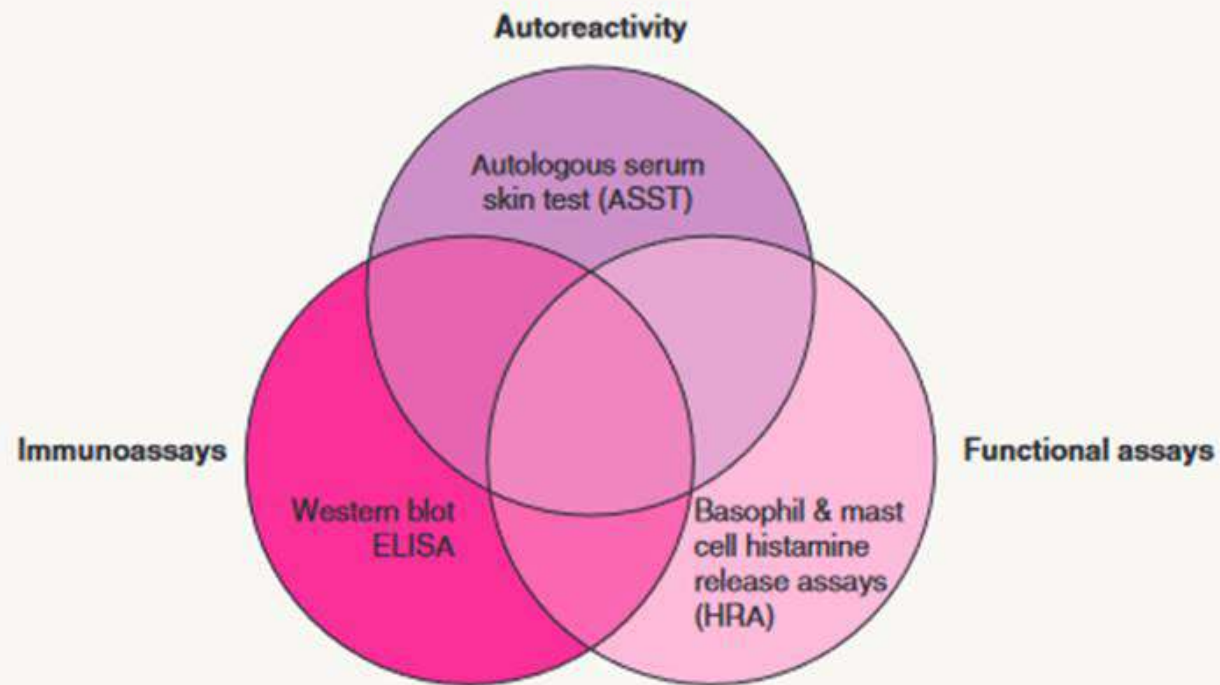




Fig. 18.17 Autologous serum skin test (ASST). At 30 minutes, a red wheal response is present at the site of an intradermal injection of autologous serum, but not at that of a normal saline control. For the test to be positive, the wheal response at the serum injection site must be at least 1.5 mm greater in diameter than the control site. This criterion is a reasonably sensitive and specific screening test for functional autoantibodies as detected by the basophil release assay. Spontaneous wheals are also present at the venipuncture site and above the serum skin test.

**OVERLAP BETWEEN A POSITIVE ASST,
IN VITRO HRA, AND IN VITRO IMMUNOASSAYS IN THE DIAGNOSIS OF AUTOIMMUNE URTICARIA**



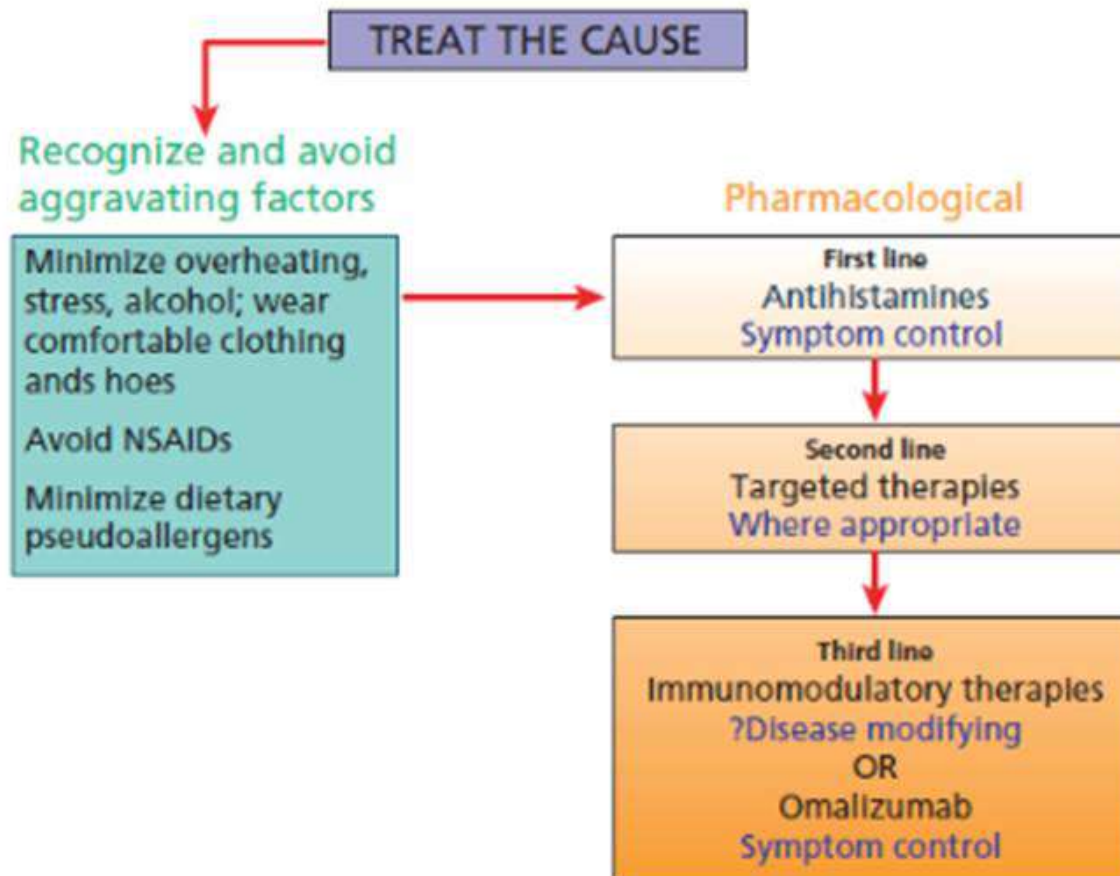
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graph TD; Treatment[Treatment] --> General[General measures]; Treatment --> Pharmacological[Pharmacological];
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Treatment

General
measures

Pharmacological

Treatment



General measures

1. Education about triggers
2. Avoidance aggravating factors
3. Stop drug
4. Decrease stress , heat , alcohol
5. Decrease food additive
6. Antipruritic , cooling lotion
7. High dose vit D

Pharmacological

1st line

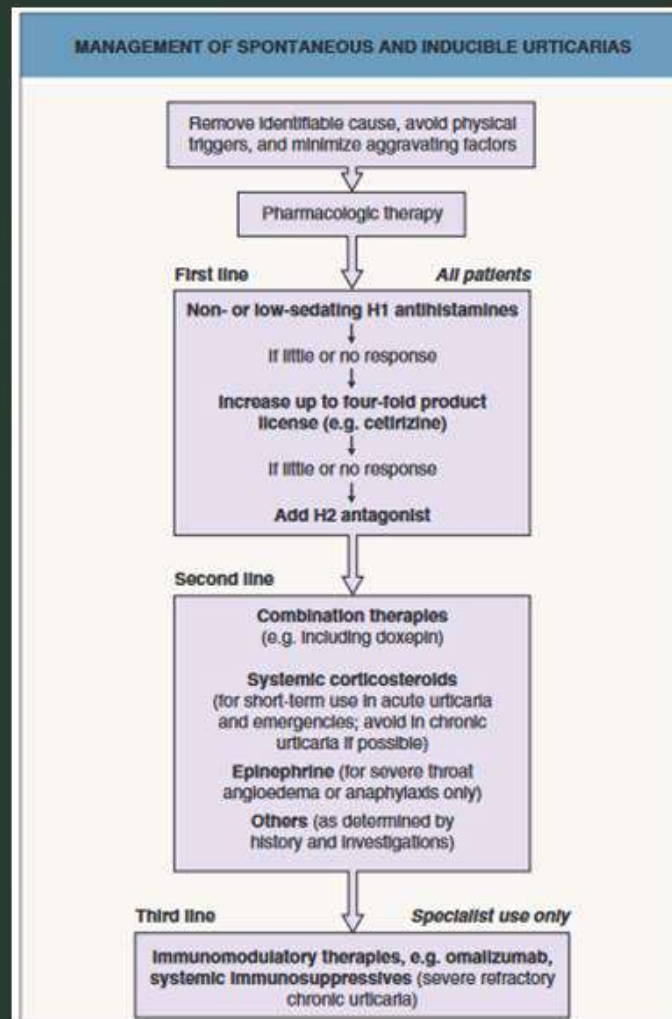
Antihistaminic
1st generation
2nd generation

2nd line

Steroid
Epinephrin
Montelukast
Colchecin
Sulfasalazin
Dapson

3rd line

Cyclosporin
omalizumab



ANTIHISTAMINES FOR URTICARIA

Rx

Class	Examples	Plasma half-life (hours)	Daily adult dose*
Classic (sedating) H ₁ antihistamines	Chlorpheniramine (1)	12–15	4 mg three times daily (up to 12 mg at night)
	Hydroxyzine (1)	20	10–25 mg three times daily (up to 75 mg at night)
	Diphenhydramine (2)	4	10–25 mg at night
	Doxepin [†] (1)	17	10–50 mg at night
Second-generation H ₁ antihistamines	Acrivastine [‡] (1)	2–4	8 mg three times daily
	Cetirizine [§] (1)	7–11	10 mg once daily
	Loratadine (1)	8–11	10 mg once daily
	Mizolastine (1)	13	10 mg once daily
Newer second-generation H ₁ antihistamines	Desloratadine (1)	19–35	5 mg once daily
	Fexofenadine (1)	17	180 mg once daily
	Levocetirizine (1)	7–10	5 mg once daily
	Rupatadine (1)	6	10 mg once daily
H ₂ antagonists [¶]	Cimetidine (1)	2	400 mg twice daily
	Ranitidine (2)	2–3	150 mg twice daily

SECOND-LINE MEDICATIONS FOR CHRONIC OR PHYSICAL URTICARIA

Generic name	Drug class	Route	Dose	Special indication/associated diseases
Prednisone (2)	Corticosteroid	Oral	0.5 mg/kg daily	Severe exacerbations (days only) – “rescue” treatment
Epinephrine (2)	Sympathomimetic	Subcutaneous, IM (self-administered)	300–500 mcg	Idiopathic or allergic angioedema of throat/anaphylaxis
Montelukast (3)	Leukotriene receptor antagonist	Oral	10 mg daily	Aspirin-sensitive urticaria, ? delayed pressure urticaria
Colchicine (3)	Neutrophil inhibitor	Oral	0.5/0.6–1.5/1.8 mg* daily	Neutrophilic infiltrates in lesional biopsy specimens or urticarial vasculitis
Sulfasalazine (3)	Aminosalicylates	Oral	2–4 g daily	Delayed pressure urticaria

